MATH329 · OPERATIONS RESEARCH

# Midterm 472 Solutions

13/04/2026 · 60 minutes · 30 marks

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## 01 Solution Q1: True / False

| # | Answer | Reason |
|---|---|---|
| 1 | **F** | The graphical method works for two variables only, not for more than three. |
| 2 | **F** | It started during **World War II**, not the first war. |
| 3 | **F** | The objective and all the constraints must be linear, not one or the other. |
| 4 | **T** | The tableau organizes the objective and the constraint equations. |
| 5 | **F** | The solution must satisfy the constraints, not "may". |
| 6 | **T** | This is one of the two requirements to start simplex. |
| 7 | **T** * | This is what the answer key shows. See the note below. |

**A note on item 7**

The key shows T. By the lecture, when no variable can enter, the current solution is optimal (**the last solution is optimal**); it does not mean the problem has no feasible solution. So by the lecture the answer is F. The key and the lecture disagree here, so confirm with the instructor.}}

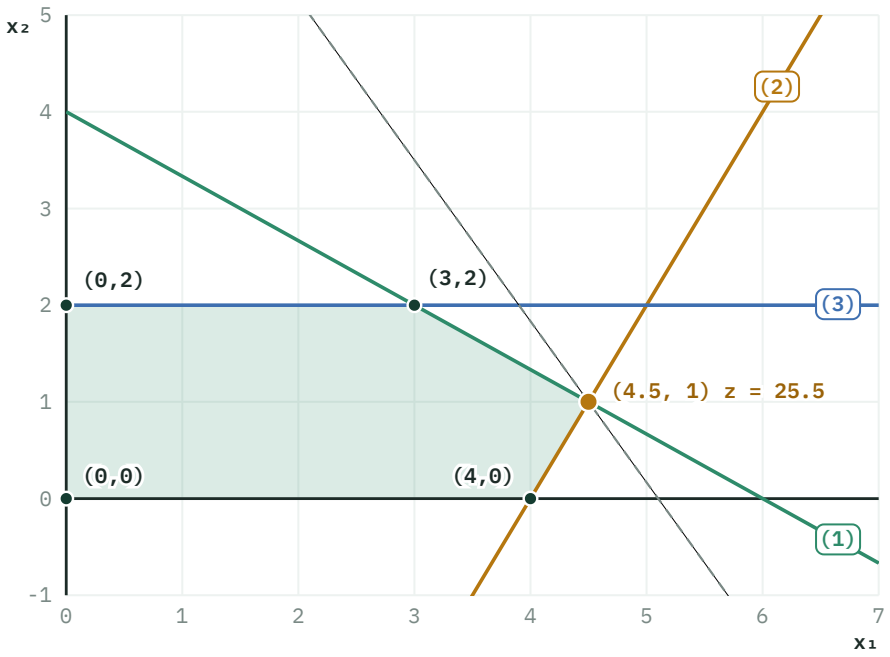
The answer to item 2 is not visible in the photo of the key; F is taken from the lecture.

## 02 Solution Q2: graphical

### 1 Prepare the constraints

| | | |
|---|---|---|
| (1) | $2x_1 + 3x_2 = 12$ | points (0, 4) and (6, 0) |
| (2) | $2x_1 - x_2 = 8$ | points (4, 0) and (0, −8) |
| (3) | $-2x_2 \geq -4$ × (−1) → | $x_2 \leq 2$ horizontal line at $x_2 = 2$ |

Testing `(0, 0)` satisfies all three, so the region is on the origin's side.



## 2 The corner points

| Corner | Where it comes from | $Z = 5x_1 + 3x_2$ |
|---|---|---|
| (0, 0) | origin | 0 |
| (4, 0) | (2) with the $x_1$ axis | 20 |
| **(4.5, 1)** | (1) with (2) | **25.5** |
| (3, 2) | (1) with (3) | 21 |
| (0, 2) | (3) with the $x_2$ axis | 6 |

(1) − (2): $4x_2 = 4 \rightarrow x_2 = 1$ $\quad$ $2x_1 - 1 = 8 \rightarrow x_1 = 4.5$

**Objective Value = 25.5 at $x_1 = 4.5$ and $x_2 = 1$**

Marks in the key: one for each of the three lines, one for the objective line, two for the solution space, one for the optimum point, and two for the answer.

# 03 Solution Q3: standard form and simplex

## a Standard form 5 M

The second constraint has a negative right-hand side, so multiply it by $-1$ first:

$$x_1 - x_2 \geq -1 \quad \times (-1) \quad \rightarrow \quad -x_1 + x_2 \leq 1$$

$$\begin{aligned}
&\text{Maximize } Z = 16x_1 + 15x_2 + 0s_1 + 0s_2 + 0s_3 \\
&\text{Subject to} \quad 40x_1 + 31x_2 + s_1 = 124 \\
&\qquad\qquad -x_1 + x_2 + s_2 = 1 \\
&\qquad\qquad x_1 + s_3 = 3 \\
&\qquad\qquad x_1, x_2, s_1, s_2, s_3 \geq 0
\end{aligned}$$

**b** **Simplex** 10 M · 2 TABLES ONLY

| Basic | $x_1$ | $x_2$ | $s_1$ | $s_2$ | $s_3$ | Sol. | Ratio |
|---|---|---|---|---|---|---|---|
| Z | −16 | −15 | 0 | 0 | 0 | 0 | |
| $s_1$ | 40 | 31 | 1 | 0 | 0 | 124 | $\frac{31}{10}$ |
| $s_2$ | −1 | 1 | 0 | 1 | 0 | 1 | – |
| $s_3$ | **1** | 0 | 0 | 0 | 1 | 3 | 3 |
| Z | 0 | −15 | 0 | 0 | 16 | 48 | |
| $s_1$ | 0 | 31 | 1 | 0 | −40 | 4 | |
| $s_2$ | 0 | 1 | 0 | 1 | 1 | 4 | |
| $x_1$ | 1 | 0 | 0 | 0 | 1 | 3 | |

The most negative entry in the `Z` row is `−16`, so $x_1$ enters. The ratios are $\frac{124}{40} = 3.1$ for the first row, none for the second row (negative coefficient), and $\frac{3}{1} = 3$ for the third. The smallest is `3`, so $s_3$ leaves and the pivot is `1`.

New $x_1$ row = $s_3$ row ÷ 1 = (1, 0, 0, 0, 1 | 3)

New Z row = Z row − (−16) × new $x_1$ row = (0, −15, 0, 0, 16 | 48)

New $s_1$ row = $s_1$ row − 40 × new $x_1$ row = (0, 31, 1, 0, −40 | 4)

New $s_2$ row = $s_2$ row − (−1) × new $x_1$ row = (0, 1, 0, 1, 1 | 4)

We stop here because the question asks for two tables only. The solution is not optimal yet, because `−15` remains in the `Z` row.